Turbulence Discussion

Questions to Ponder... and Answer

For the turbulence model you are discussing...

- How are the $\overline{u_i u_j}$ terms in the momentum equations approximated?
- What turbulence quantity or quantities are calculated by using a PDE?
- How is/are the equation(s) for that quantity(ies) derived? From physics? Or heuristically? In either case, what is reasoning behind the choices made in the derivation?
- What terms in the turbulence PDE are calculated exactly and which are modeled?
- What flows are used to calibrate the constants in the model?
- For what flows does the turbulence model perform well (= match experimental results)?
- For what flows does the model perform poorly?
- Based on cases in the papers you have read, what strengths and weaknesses can you note in other turbulence models that have been used for comparison?